

Weather Tracker App

An interactive Streamlit and Pandas dashboard localized for the climate patterns of Bahrain.



[GitHub Project Repository Demo](#)

Bahrain Climate Observations

Why Localized Tracking Matters

Bahrain features a extreme dual-season desert climate with intense heat variations. Capturing precise local environmental metrics requires high-fidelity recording systems.

- ◆ **Summer Extremes:** High-humidity Shamal wind systems ranging upwards of 45°C.
- ◆ **Mild Winters:** Brief cooling periods dropping down to comfortable ranges of 14°C.
- ◆ **The Challenge:** Building a lightweight, fast, and structured platform that logs, searches, and monitors these distinct shifts over time.



Modular System Architecture

main_app.py

Acts as the clean, modern user interface layer developed with the Streamlit framework.

- ◆ Loads the page sidebar navigation dynamically using `st.sidebar.radio`.
- ◆ Captures structured inputs via isolated form blocks to prevent constant re-rendering.
- ◆ Executes validation filters before invoking backend actions.

weather_utils.py

Acts as the dedicated calculations engine executing all logical operations using Pandas.

- ◆ Reads from and appends directly to `weather_observations.csv`.
- ◆ Aggregates observations to produce averages, minimums, and maximum values.
- ◆ Runs efficient data filters to locate entries matching target dates.

Key Dashboard Modules



1. Record Input

Simple form component collecting date string, temperature, sky condition slider ranges, and wind speeds safely without page reset bugs.



2. View Statistics

Extracts numeric averages, extreme outliers, and maps dominant climate trends across logs using automated calculations.



3. Search by Date

Locates historical entries instantly matching selected calendar dates and displays formatted observation datasets to users.

Live Demonstration

Thank you for your attention. Let's head over to the terminal and examine the active Streamlit app interfaces.

Ready to address any conceptual queries regarding Python structure, Pandas processing, or Streamlit layout designs.

 github.com/username/weather-tracker-app

Image Sources



<https://media.istockphoto.com/id/527616137/photo/skyline-of-manama-city-bahrain-middle-east.jpg?s=612x612&w=0&k=20&c=DKjGgeOrZ4tS3dYAGsAeiTpqNCFRhxOAs0zFHQX3vjl=>

Source: www.istockphoto.com